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Palaeomagnetism of a sequence of red beds of the Middle and Upper Sections of Pagnazo Group (Argentina) and the correlation of Upper Palaeozoic–Lower Mesozoic rocks

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Summary. Palaeomagnetic data from 182 hand samples collected in a rock sequence of about 620-m of red beds of Late Palaeozoic to Early Triassic age exposed in north-western Argentina (30.3° S 67.7° W), are given.

After cleaning, the majority of the Upper Palaeozoic samples (Middle Section of Paganzo Group) show reversed polarity and yield a palaeomagnetic pole at 78° S 249° E ($\alpha_{95} = 3°$). They also record a polarity transition which we have correlated with the Middle Permian Quebrada del Pimiento Normal Event. The position of the palaeomagnetic pole and the K–Ar age of a basalatic sill at the base of the sequence support this correlation.

Stable remanent magnetization has been isolated in the majority of samples from the Upper Section of the Paganzo Group; it is predominantly reversed and reveals three normal events and also three geomagnetic excursions suggesting an Illawarra Zone age (post Kiaman, Late Tatarian–Early Scythian). The palaeomagnetic pole of the reversely magnetized samples is located at 75° S 285° E ($\alpha_{95} = 13°$).

The red beds involved in this study are correlated with red beds from the Corumbataí Formation (State of Paraná, Brazil) and with igneous rocks from the Quebrada del Pimiento Formation (Province of Mendoza, Argentina).

The South American Middle and Upper Permian, Upper Permian–Lower Triassic, Lower, Middle and Upper Triassic and Middle Jurassic palaeomagnetic poles reflect a quasistatic period with mean pole at 82° S 244° E, ($\alpha_{95} = 4°$) which followed the South American Late Palaeozoic polar shift.

1 Introduction

The direction and polarity of the stable remanence of sequences of Upper Palaeozoic–Lower Mesozoic rocks of South America and Africa have been used to deduce the magnetic stratigraphy of these successions of rocks (Valencio 1972, 1975; Valencio, Rocha Campos & Pacca 1975a; Rocha Campos, Valencio & Pacca 1976). This procedure is only effective when it is based on a detailed and systematic collection of samples from thick successions of rocks.

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In such cases, variations in declination and inclination of the stable remanence through a stratigraphic succession can be used to correlate different successions.

In north-western Argentina, outcrops of large exposures of continental sedimentary deposits and igneous rocks comprising the Paganzo Group have been assigned a Late Palaeozoic to Early Mesozoic age. The palaeomagnetic study of rock sequences of the Paganzo Group has been included in the Argentinian Programme for the Upper Palaeozoic of South America Project of the International Geological Correlation Programme. The results of the study of a rock sequence of the Paganzo Group are presented and discussed in this paper.

Palaeomagnetic data of igneous rocks and red beds of the Paganzo Group have been reported by Creer (1965, 1970a), Creer, Embleton & Valencio (1969, 1970), Embleton (1970a, b) and Thompson & Mitchell (1972). These data have been critically analysed by Valencio (1972). In this paper a discussion of the radiometric and geologic ages of the Paganzo rocks used in the palaeomagnetic studies is given.

Thompson (1972) studied 71 hand samples of red beds from the Middle Section (La Colina Formation) and from the Upper Section (Amaná Formation) of the Paganzo Group collected near Paganzo Village. We present here the results of a palaeomagnetic study of 182 hand samples collected from a rock sequence of about 620-m of red beds of the Middle and Upper Sections of the Paganzo Group exposured near Paganzo Village. Some of these samples were collected from the site sampled by Thompson but our new collection is more complete. Our field work also included a detailed study of the geology of the site.

The magnetic stratigraphy defined for the Middle and Upper Sections of the Paganzo Group exposed in Paganzo Village provides a reference profile which may be used for geological correlation within the Paganzo Basin. This magnetic stratigraphy may be also used for the geological correlation of Paganzo rocks with rock sequences of other basins all over the world.

2 Available data

2.1 GEOLOGICAL EVIDENCE

In the southern part of the Province of La Rioja, Argentina (Fig. 1), are exposed continental deposits and igneous rocks included in the Paganzo Group. These rocks were formed in Late Palaeozoic and Early Mesozoic time and have been classified in three sections: a Lower Section (Paganzo I), a Middle Section (Paganzo II) and an Upper Section (Paganzo III). The sequence of Paganzo rocks included in this study (Paganzo Village, 30.3° S 67.7° W, Fig. 1) have been further divided into three formations named in chronological order: Lagares Formation, La Colina Formation and Amaná Formation which were correlated, respectively, with the three sections mentioned before (Azcuy & Morelli 1970). Deposition of sediments of the Lower Section began after the Acadian–Breton phase of the Carboniferous Geosyncline; conglomerate layers at the base of the section are followed by light coloured pelites, arkoses and sandstones with intercalations of conglomerates and carbonaceous clays. Middle Section sediments consist of fine-grained red sandstones and cross-bedded siltstones. Basaltic flows and sills are a characteristic feature of this Middle Section. The type of terrestrial deposits included in the Upper Section differs from area to area.

It has been indicated that the Upper Section sediments rest unconformably on the Middle Section rocks. In the Paganzo rock sequence exposed near Paganzo Village involved in this study that unconformity is not evident. However it is accepted that an unknown sequence of the youngest sediments of the Middle Section is not represented in the area (J. Morelli & C. Azcuy, private communication). The correlation between exposures in different areas of the Paganzo Basin is extremely difficult.

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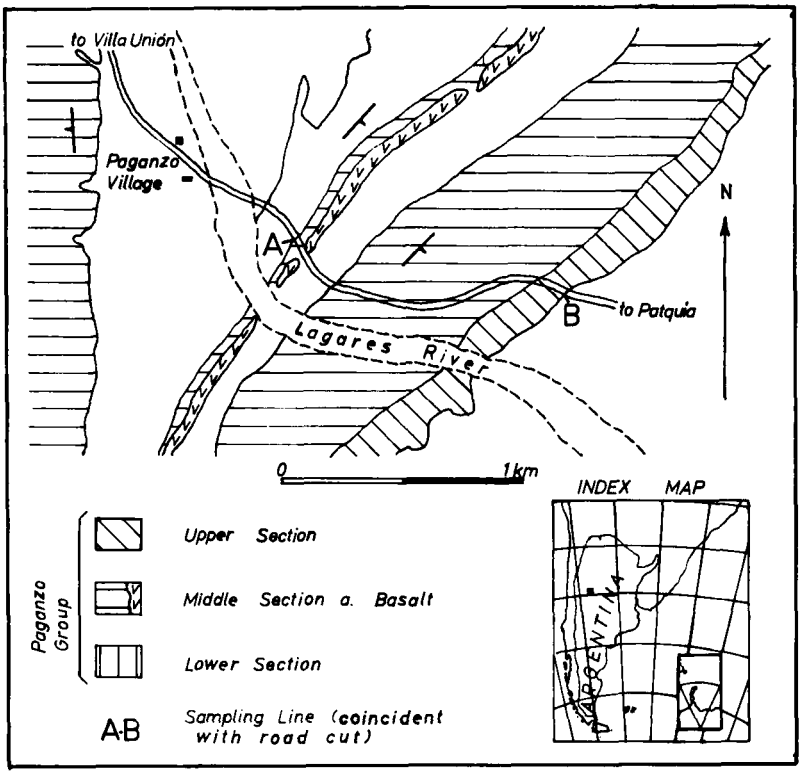


Figure 1. Schematic geological map of Paganzo Basin near Paganzo Village (Azcuy & Morelli 1970) showing sampling sites for the palaeomagnetic study.

During the Tertiary Andean orogenesis the Paganzo rocks were folded and faulted along reactivated lines of weakness in the Lower Palaeozoic and Pre-Cambrian basement rocks.

It is generally agreed that the deposition of sediments of the Lower Section of the Paganzo Group started in Carboniferous time. Valencio (1972) and Valencio & Mitchell (1972) suggested a Late Carboniferous age for the red beds of the Middle Section exposed in Huaco and Los Colorados on the basis of their stable magnetic remanences. Thompson & Mitchell (1972) obtained a K–Ar age of 266 ± 5 Myr for a basaltic sill exposed close to the base of the Middle Section red beds in Paganzo Village; these authors, assuming that it was a lava flow, regarded this age as low due to argon loss and preferred to conclude that it was stratigraphically equivalent with a basaltic lava flow exposed near the base of the Middle Section of the Paganzo Group at Las Torres which they had dated at 295 ± 5 Myr. Valencio (1972) questioned this interpretation and argued so that the igneous units at Las Torres and Paganzo Village could have originated from different and independent magmatic episodes. This view is supported by field work reported in this paper (Section 2,4) and by K–Ar age determinations carried out on two new samples of the Paganzo Village basaltic sill which give ages coherent with those obtained by Thompson & Mitchell (Section 2.2).

On the basis of the position of the palaeomagnetic pole computed by Thompson (1972) for the Middle Section exposures in Paganzo Village, Valencio (1972) suggested a magnetic age younger than Late Carboniferous but older than Middle Permian and probably close to the Permian–Carboniferous boundary for these rocks. This interpretation is modified by the results obtained in the present palaeomagnetic study.

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The age of Upper Section deposits is uncertain although they are assigned to the Triassic because sedimentary rocks of Middle Triassic age rest on the Upper Section fixing an age older than Middle Triassic for it.

2.2 POTASSIUM–ARGON AGE DETERMINATIONS

Whole K–Ar age determinations were carried out on two hand samples collected from exposures of the basaltic sill near Paganzo Village by Embleton (1970a). The study was carried out on sample numbers 2043 and 2044 of Embleton's collection in the Instituto de Geocronología y Geología Isotópica (Universidad de Buenos Aires). A petrological study indicated that both samples are fresh without evident alteration. The equipment and techniques used are those given by Linares & Valencio (1975). The K–Ar ages are shown in Table 1. Sample 2043 yields an age coincident with the K–Ar ages obtained for a sample from the same igneous unit by Thompson & Mitchell (1972). The ages obtained from sample 2044 are considered to be anomalously affected by atmospheric argon. The mean of the four reliable K–Ar ages for the basaltic sill of the Middle Section of the Paganzo Group near Paganzo Village is 266 ± 7 Myr suggesting a Middle Permian age.

Table 1. K–Ar ages for the basaltic sill of the Middle Section of Paganzo Group (Paganzo Village).

Sample number	K (%)	Ar^{40}_{rad} (10^{-10} mol/g)	Ar^{40}_{atm} (%)	Age (Myr)
2043	0.44	2.24	22.4	266 ± 15
2044	0.15	0.835	85.8	290 ± 25
		0.789	83.8	274 ± 25

2.3 THE PALAEOMAGNETIC STUDY

Red sediments from the Middle (La Colina Formation) and Upper Sections (Amaná Formation) of the Paganzo Group were collected near Paganzo Village (Fig. 1, Azcuy & Morelli 1970) from the eastern limb of an anticline along the route 26 between Paganzo Village and Patquía. Sampling started from the lowest exposures of the Middle Section in the area and extended up to the highest exposures of the Upper Section covering a sequence of about 620 m. Wherever possible two hand samples were collected at stratigraphic intervals of 2 m from different but contiguous stratigraphic levels. Samples were also collected from the baked sediments above and below the basaltic sill present near the base of the sequence. The palaeomagnetic study of this basalt was carried out by Embleton (1970a).

One hundred and thirty-two and 50 hand samples of sedimentary rocks were collected respectively from the Middle and Upper Sections of the Paganzo Group. Hand samples were orientated using both magnetic and sun compasses. Thirty-eight observations of the bedding plane were carried out (27 in Middle Section beds and 11 in Upper Section beds).

More than 550 cylinders of 2.5-cm diameter and height were cut. Two spinner magnetometers (one a Digico and one made in the Universidad de Buenos Aires) were used to measure the natural remanence of cylinders. Detailed thermal cleaning was used to isolate the stable remanence of the red beds; one pilot cylinder from each hand samples was heated to progressively increasing temperatures in steps of about 100°C up to the Curie temperature and then cooled in magnetic field-free space. The criterion adopted for the choice of the optimum cleaning temperature was that the direction of the remanence of the pilot cylinder should exhibit no further change in response to higher temperatures. At least a second

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Table 2. Summary of bedding plane observations of the Paganzo Group sequence from the eastern limb of the anticline situated near the Paganzo Village.

Paganzo Group	Bedding plane				
	N	Strike (°)	Dip (°)	α_{95} (°)	k
Upper Section	11	53	52	4	108
Middle Section	19	50	50	3	107

cylinder from the same hand samples was submitted to the optimum temperature so as to improve the definition of the stable RM. The stable remanence of the baked sedimentary rocks was isolated by a similar detailed alternating field demagnetization. The statistical method of Fisher (1953) was used to compute the mean directions of NRM and cleaned RM of each hand sample having assigned unit weight to the direction of each of its component cylinders.

Sample-mean directions of NRM and cleaned RM were referred to the palaeohorizontal (bedding plane correction). However, the fold test could not be applied because all the samples come from one area where the dip of the strata is uniform. Table 2 summarizes the bedding plane observations carried out in the sequences of the Middle and Upper Sections; the measurements made on cross bedded strata were rejected.

Five hand samples were destroyed during transportation or cutting. It was not possible to isolate stable remanence in five samples from the Upper Section. The palaeomagnetic measurements from the remaining 172 hand samples of the Paganzo Group are reported below.

2.3.1 Middle Section

The mean directions, after bedding plane correction, of NRM and cleaned RM of each hand sample from the Middle Section of the Paganzo Group are shown in Fig. 2(a) and (b). Note that all samples but one yield mean directions with positive inclinations. NRM directions lie close to the centre of the stereographic net and are reasonably well grouped. The mean direction, referred to the palaeohorizontal, of the NRM of the 129 hand samples is given in Table 3.

A weak secondary RM is present in many of the red beds but was easily destroyed by the thermal treatment. The optimum demagnetizing temperature varied between 250 and 400°C at which more than 50 per cent of the natural intensity was retained. Demagnetization of pilot cylinders suggests that the carrier of stable remanence is haematite.

Note that the normally magnetized sample retains its normal polarization after cleaning. In Table 3 it is shown that the sample-mean directions are better grouped after cleaning ($k = 15$) than before ($k = 10$) and this is also evident when Fig. 2(a) and (b) are compared.

2.3.2 Upper Section

Sample-mean directions of NRM and of cleaned RM, referred to the palaeohorizontal, are shown in Fig. 3(a) and (b).

A soft secondary component of remanence, present in the majority of the red beds, was destroyed by thermal cleaning and a stable remanence was isolated in the majority of the samples. The optimum demagnetizing temperature varied between 250 and 350°C at which more than 0.45 of the NRM was retained. The directions are more tightly grouped after

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Table 3. Summary of palaeomagnetic results of the Paganzo Group succession from the eastern limb of the anticline situated near the Paganzo Village.

Paganzo Group	Natural remanent magnetization referred to the palaeohorizontal					Cleaned remanent magnetization referred to the palaeohorizontal					Palaeomagnetic pole			
	N	D (°)	I (°)	α_{95} (°)	k	N	D (°)	I (°)	α_{95} (°)	k	Lat South (°)	Long East (°)	d_X (°)	d_ψ (°)
Upper Section	43	62	78	15	3	43	155	67	12	4	63	329	20	16
Middle Section	129	210	76	4	10	129	189	59	3	15	78	257	4	3

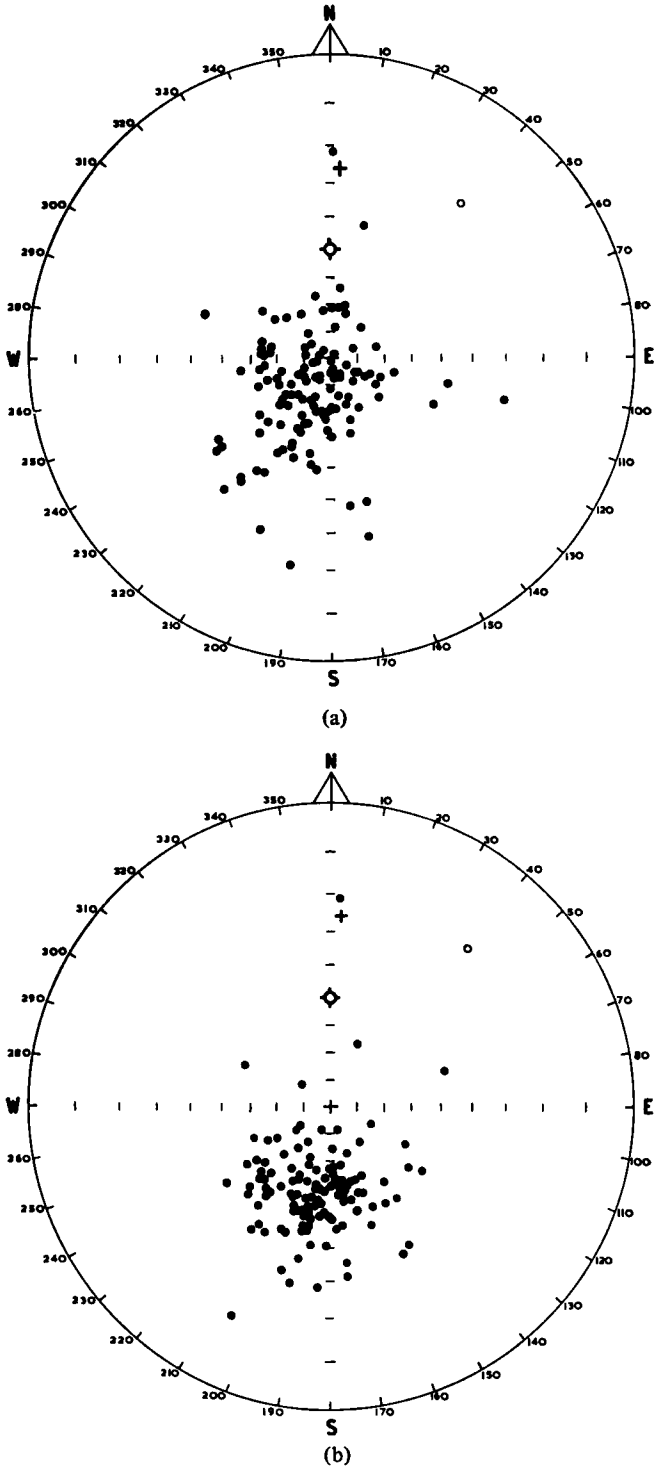


Figure 2. Paganzo Village; Middle Section of the Paganzo Group. Directions of (a) NRM and (b) cleaned RM of hand samples referred to the palaeohorizontal. Solid symbols indicate downward dipping directions. The directions of the present geomagnetic field and of the axial dipole are shown by + and ⊹ respectively.

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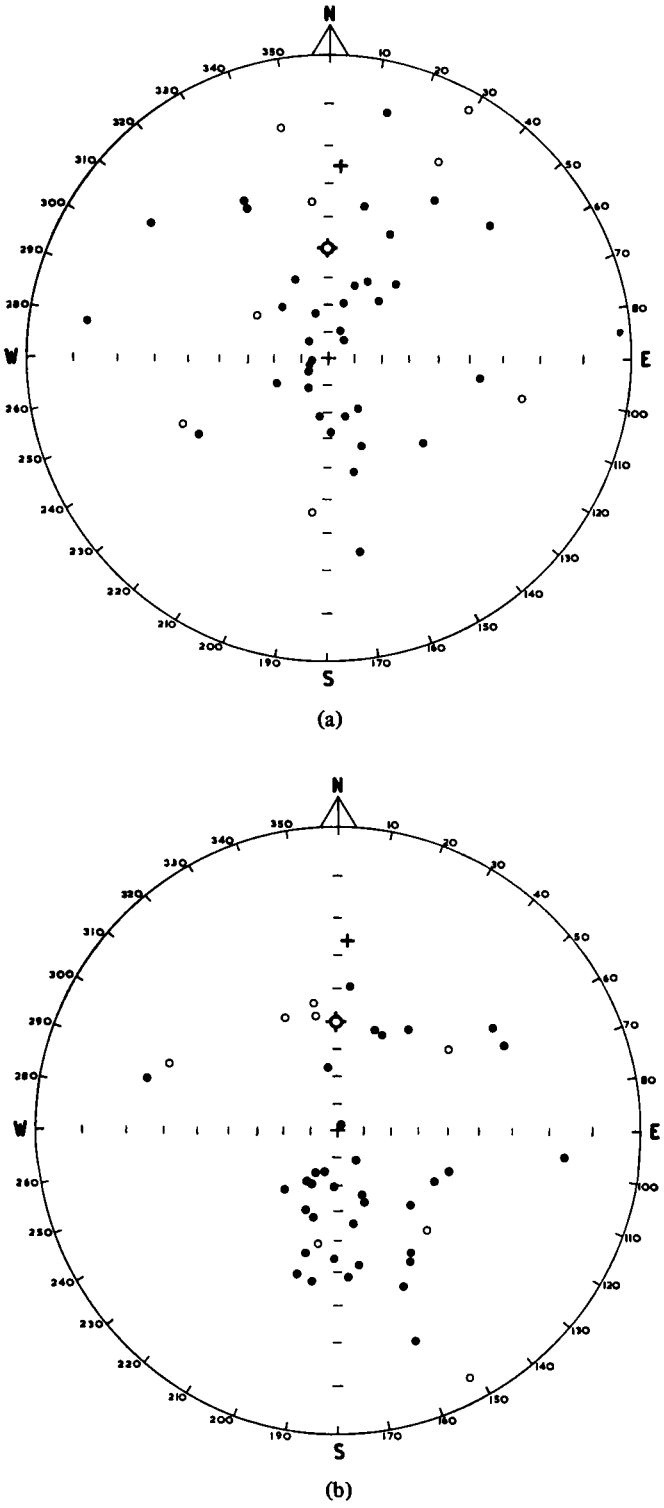


Figure 3. Paganzo Village; Upper Section of the Paganzo Group. Direction of (a) NRM and (b) cleaned RM of hand samples referred to the palaeohorizontal. The key to the symbols is as given in Fig. 2.

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thermal treatment (Fig. 3 and Table 3). However, some of the cleaned directions do not lie in the main group of reversely polarized directions. An explanation of this is given in Section 3.

3 Interpretation of results

The sequence of sediments and igneous rocks of the Paganzo Group involved in this study is about 620 m thick; but for geological reasons, it was only possible to cover adequately the upper 300-m. The magnetic stratigraphy consisting of logs of sample-mean values of cleaned declination (D) and inclination (I) is plotted alongside the lithologic log in Fig. 4. The character of the variations in D and I above the level of samples 6203–6204 differs from that below it where D and I undergo relatively small oscillations around 190° and 55° respectively. This quiet behaviour of an apparently stably oscillating reversely polarized geomagnetic field holds throughout the Middle Section except at the levels of samples 6158–6164 where northerly declinations and negative inclinations are recorded. In contrast, above the level of samples 6203–6204, the geomagnetic field behaved quietly only between the levels of samples 6214 and 6225. Elsewhere, frequent reversals of both declination and inclination are recorded and we associate these with short-lived polarity transitions or with geomagnetic excursions.

It is generally agreed that the Middle Section of the Paganzo Group should be assigned to the late Palaeozoic. The age of the Upper Section of the Paganzo Group is uncertain although it is generally placed in the Triassic (Section 2.1). Our collection of Middle Section red sediments runs upwards from a stratigraphic level just below a basaltic sill, Fig. 4(a), exposed at Paganzo Village (Section 2.1). We accept the K–Ar age 266 ± 7 Myr (Section 2.2) obtained for this sill as reliable. Thus the age of the stratigraphically lowest sediments of the Middle Section of the Paganzo Group involved in this study must be older than Middle Permian. How much older we do not know but a late Early Permian or early Middle Permian age seems reasonable to us. This indicates a Kiaman age, older than the Quebrada del Pimiento Normal Magnetic Event (Valencio & Vilas 1972) and younger than the Oak Creek Normal Magnetic Event (McElhinny 1969) and this is consistent with the reversed magnetic remanence isolated in samples collected from the stratigraphic level of samples 6069–6070 to that of samples 6155–6156. The short polarity event recorded in samples 6157–6164 (see below, Fig. 4) could therefore be the Middle Permian Quebrada del Pimiento Normal Magnetic Event at 263 My as defined by Valencio & Mitchell (1972) though we cannot rule out the possibility that it might be a new normal event not previously reported. In this case the real Quebrada del Pimiento Event would have to have occurred in time represented by one of the four gaps in the sequence. We prefer the first interpretation because it is simpler and it does not oblige us to introduce another normal event within the Kiaman Magnetic Interval.

The transition (unconformity) between the Middle Section (Middle Permian, Kiaman age) and the Upper Section (Triassic, post Kiaman age) is not detectable in the exposures near Paganzo Village. The Kiaman Magnetic Interval ended in Late Tatarian or Early Scythian time. This interval of predominantly reversed polarity was then interrupted by a period of frequent polarity changes (Illawarra Zone). We have seen (Section 2.3) that within the sequence of rocks involved in this study is defined a net change in the behaviour of the geomagnetic field at the level of samples 6203–6204. It could be the transition between the Kiaman Interval and the Illawarra Zone. Therefore, the transition between the Middle Section (Kiaman) and the Upper Section (Illawarra) can be inferred on the basis of stable RM.

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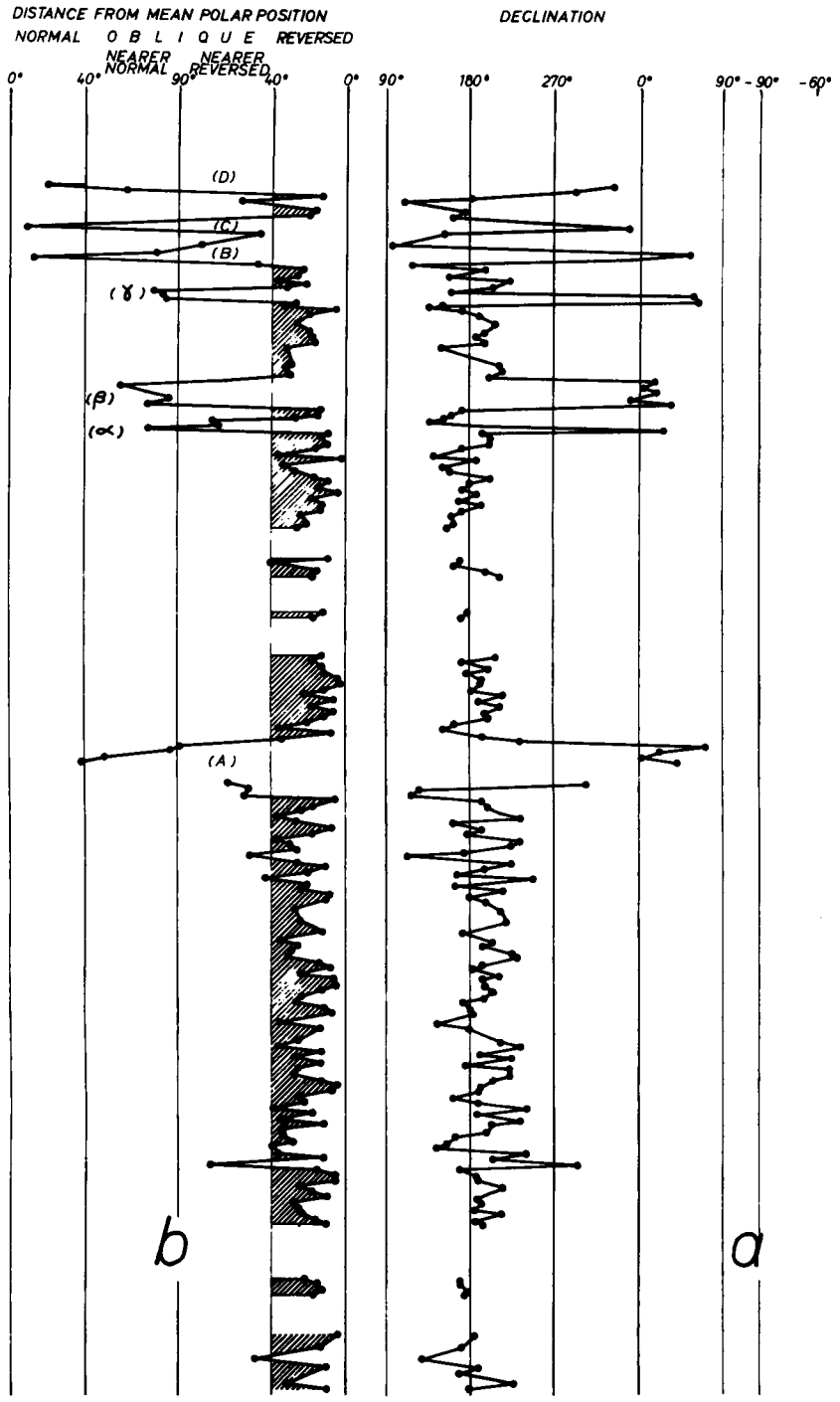
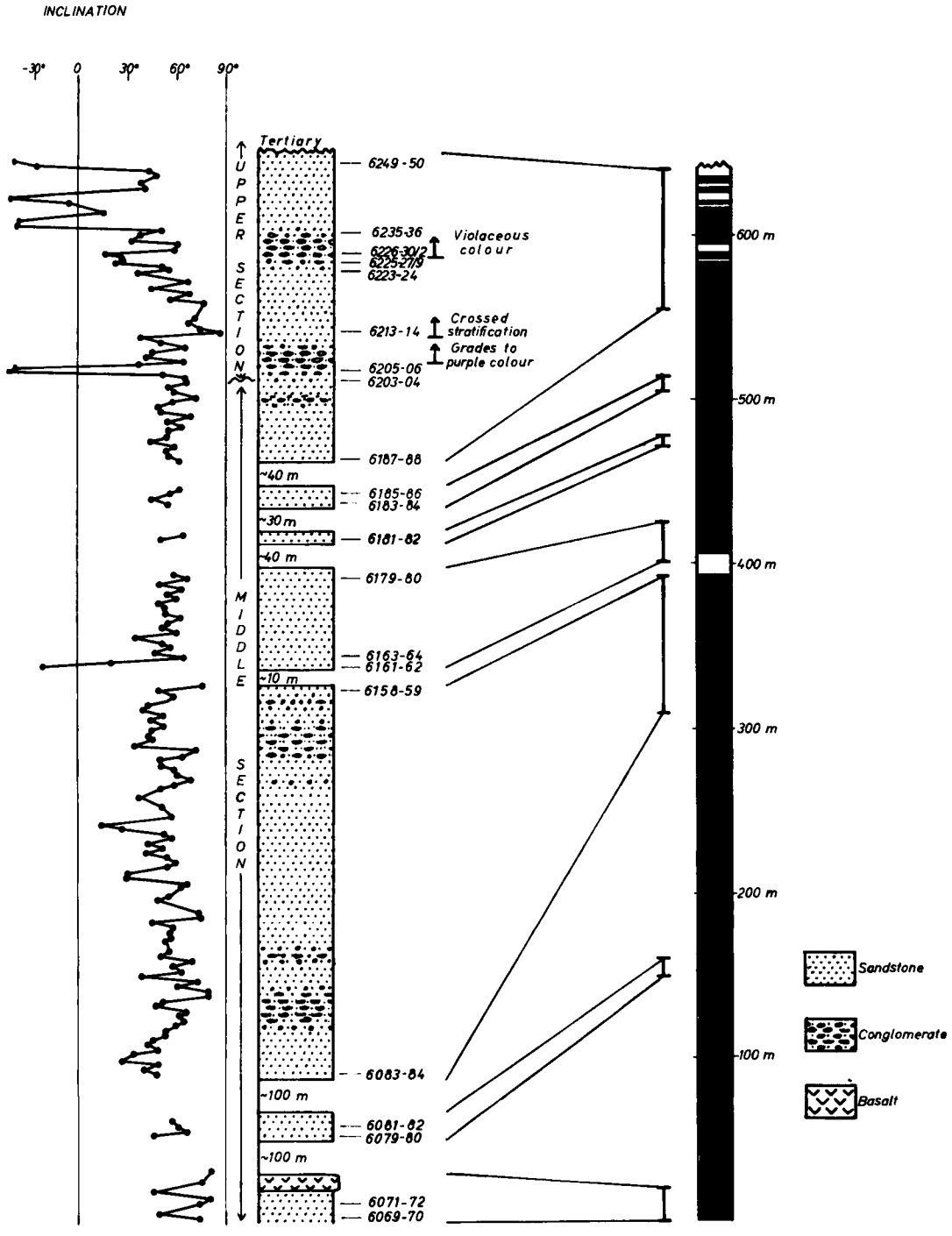


Figure 4. Declination and inclination of stable remanence and the angle between the VGP for each hand sample and the palaeomagnetic pole through the Middle and Upper Sections of the Paganzo Group. The schematic lithologic column is also illustrated.

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To proceed further, palaeomagnetic poles were calculated for the Upper and Middle Sections by meaning the sample-mean VGP's. Next, we rejected the VGP's situated more than 60° from the respective palaeomagnetic poles and new palaeomagnetic poles were then computed. The process was repeated until a population of VGP's was obtained all within 40° of the palaeomagnetic pole for each section. These new populations of VGP's give palaeomagnetic poles for the Middle and Upper Sections of the Paganzo Group at 78° S 249° E ($N = 120$, $k = 15$, $\alpha_{95} = 3°$, P_7) and 83° S 317° E ($N = 26$, $k = 12$, $\alpha_{95} = 8.5°$), respectively. The rejected VGP's are classified as oblique. McElhinny, Embleton & Wellman (1974) also used a 40° cut-off.

We interpret an observed sequence of oblique VGP's as indicating a short polarity event within the reversed succession of rocks if in the limit, at least one VGP lies within 40° of the normal palaeomagnetic pole. Examples are given by samples 6157–6164 (event A) and by samples 6235–6250 (events B, C and D) – Fig. 4(b). The other sequences of oblique VGP's, namely from samples 6203–6213 and 6229–6232, are interpreted as records of geomagnetic excursions, labelled α, β and γ in Fig. 4(b).

The position of the palaeomagnetic pole we have computed for the Middle Section of the Paganzo Group exposures near Paganzo Village (P_7, Fig. 5) supersedes that presented for rocks of the same Middle Section (La Colina Formation) and locality by Thompson (1972), because it is based on a more detailed and numerically larger collection of samples. The new position of P_7 is closer to the Middle Permian palaeomagnetic poles P_5 (Valencio 1969) and P_6 (Vilas 1969) than the earlier position of Thompson which is represented in Fig. 5 by a dotted circle. This supports the Middle Permian age we suggested for part of the Middle Section of the Paganzo Group sequence. The lower stratigraphic levels of the sequence should then be assigned to the early Middle Permian or late Lower Permian. The new age here suggested for the exposures of the Middle Section near Paganzo Village supersedes that given for these rocks by Valencio (1972) on the basis of the palaeomagnetic study carried

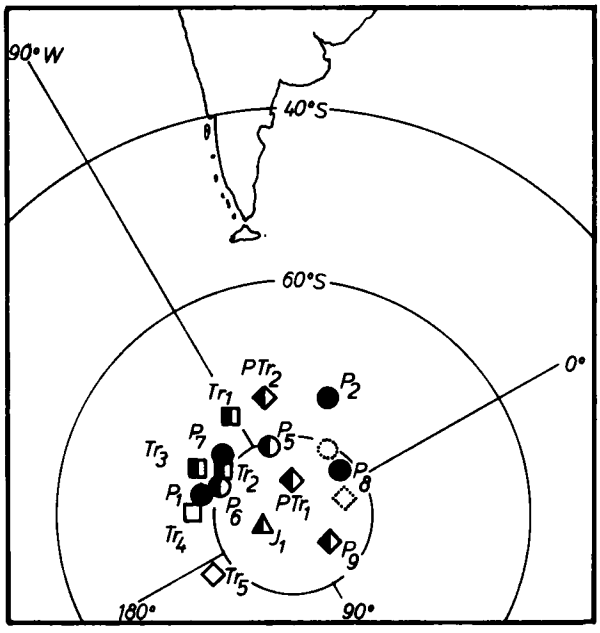


Figure 5. Late Palaeozoic and early Mesozoic palaeomagnetic poles for South America: ○, Middle and late Permian; ◇, late Tatarian–early Scythian; □, Triassic and △, Middle Jurassic. References in the text. Solid symbols reversed remanence; open symbols normal remanence.

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out by Thompson (1972). Fig. 5 also shows the positions of the palaeomagnetic poles for the Middle Section exposures at Los Colorados (P_2 and P_8, Thompson 1972). It seems to us that P_8 may be of the same age as P_7 and that P_2 may be older. The position of the palaeomagnetic pole for the Late Permian Pillahuinco Group (P_1, Creer *et al.* 1970) is also shown in this figure. The Middle and Late Permian palaeomagnetic poles for South America are well grouped with a mean at 82° S 254° E ($N = 5$, $k = 108$; $\alpha_{95} = 7°$). If we include P_2 in this group the mean becomes 82° S 270° E ($N = 6$; $k = 82$; $\alpha_{95} = 7°$) and the decrease in k suggests that P_2 does not really belong to this Permian group and supports an older age for it.

In order to remove the effects of the normal polarity events and the geomagnetic excursions we have used the cleaned remanence of samples 6213–6225, collected from horizons magnetized when the field was in a steady state of reversed polarity, to compute our palaeomagnetic pole (PTr_2) for the Upper Section of the Paganzo Group. This is located at 75° S 285° E ($N = 12$, $k = 12$, $\alpha_{95} = 13°$). We consider this to be a more reliable procedure than to use all the VGP's located within 40° of the mean pole position. This palaeomagnetic pole position is preferred to that previously computed for sediments from the Upper Section at the Paganzo by Thompson (1972) (dotted diamond, Fig. 5); firstly because a larger number of samples has been studied and secondly because the old pole was computed using all samples with stable remanence ($N = 7$) without taking into consideration whether they were magnetized by the steady reversed geomagnetic field or not. The position of PTr_2 is close to the Upper Permian–Lower Triassic palaeomagnetic pole for the Corumbataí Formation from Brazil (PTr_1, Valencio *et al.* 1975a) (Fig. 5). The cleaned magnetic remanence of the samples from the Corumbataí Formation used to compute pole PTr_1 defined three reversals of the geomagnetic field. Therefore, the stable RM strongly suggests a correlation between the sediments of the upper Section of Paganzo Group from Paganzo Village and the Corumbataí red beds.

The pole obtained by Creer *et al.* (1970) for the Upper Section of Paganzo Group exposed at Villa Unión (Tr_5, Fig. 5) should also be assigned to the Permo–Triassic though it must be noted these rocks were exclusively normally polarized.

Thompson (1972) obtained a palaeomagnetic pole for red beds of the Paganzo Group collected in Río Chaschuil (P_9, Fig. 5). He assigned these sediments to the Upper Section (Amaná Formation). On the other hand Turner (1967), in his geological study of the Rio Chaschuil area, included these sediments in the Middle Section and Azcuy & Morelli (1970) agree with this interpretation. We prefer temporarily to assign the Rio Chaschuil red beds to the Permo–Triassic because Thompson encountered both normally and reversely polarized stable remanence in the few samples collected there. However, we can not rule out the possibility of a Middle Permian age for them.

Fig. 5 shows the positions of the Triassic and Jurassic palaeomagnetic poles so far reported for South America (Tr_1, 204 ± 3 Myr, Valencio 1969; Valencio & Mitchell 1972; Tr_2, 224 ± 5 Myr, and Tr_3, 232 ± 4 Myr, Valencio, Mendía & Vilas 1975b; Tr_4, Valencio & Vilas 1972, and J_1, 166 ± 5 Myr, Vilas 1974). It can be seen that the Middle and Upper Permian, the Upper Permian–Lower Triassic (diamonds), the Lower, Middle and Upper Triassic (squares) and the Jurassic (triangle) palaeomagnetic poles for South America form a single group. Since the various studies were made on rocks collected from a wide geographical area, it follows: (i) that the Middle Permian–Jurassic geomagnetic field was predominantly dipolar and geocentric in South America, (ii) that no significant relative movements occurred between the different sampling areas from Middle Permian–Jurassic times, and (iii) that South America had a quasistatic interval from Middle Permian (or earlier) to Middle Jurassic time (Valencio 1973). The mean position of the Middle Permian–Middle Jurassic time group for South America is: 82° S 244° E (*SAP–J*; $N = 14$, $k = 90$,

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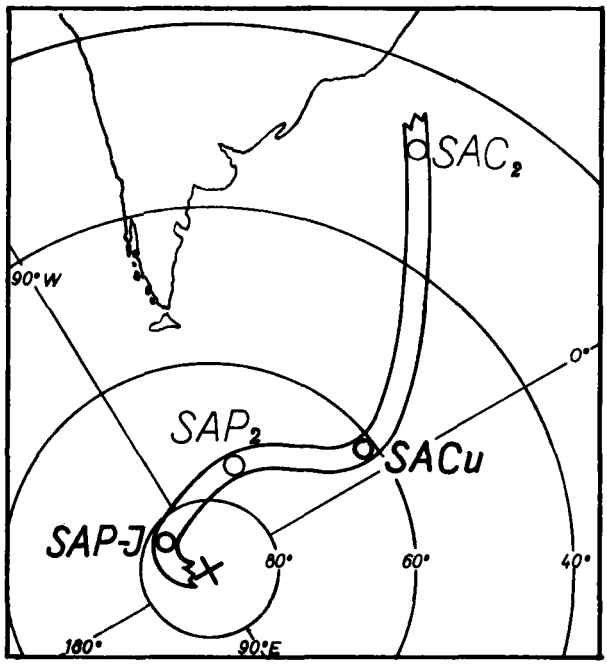


Figure 6. Late Palaeozoic section of the polar wandering curve for South America. It is defined by two mean polar positions (heavier symbols) and two palaeomagnetic poles. References in the text.

$\alpha_{95} = 4°$). We have not included P_2 in this mean. The palaeomagnetic pole positions for the Upper Section of the Paganzo Group (PTr_2, Tr_5 and P_9) lie furthest from this mean position (Fig. 5) suggesting that the non-dipolar, non-axial components of the unsteady geomagnetic field have not been totally averaged out.

Fig. 6 shows the late Palaeozoic to early Mesozoic part of the polar wandering curve for South America. SAC_2 is the palaeomagnetic pole for the Moscovian Taiguati Formation; SACu is the mean position for the age group formed by the Upper Carboniferous palaeomagnetic poles (62° S 353° E, $N = 5$, $\alpha_{95} = 4°$, Valencio *et al.* 1975a); SAP_2 is the palaeomagnetic pole for the middle levels of the Middle Section of Paganzo Group exposed in Los Colorados (Thompson 1972) and *SAP–J* is the mean position for the time group formed by the Middle Permian–Middle Jurassic palaeomagnetic poles presented in this paper. This polar path is similar to those presented in our recent papers. However it should be noted that the part of the curve between *SACu* and *SAP–J* is now located by pole SAP_2. This is the only palaeomagnetic pole belonging to the former Permo–Carboniferous age group (*SAPC*, Valencio 1973) which has not been transferred to the *SAP–J* group. Fig. 6 shows that the Middle Permian–Middle Jurassic quasistatic period (*SAP–J*) put an end to the South American late Palaeozoic polar shift.

The comparison of palaeomagnetic data obtained for a polarity transition recorded at different parts of the globe should allow us to describe the behaviour of the transitional geomagnetic field. In particular, the comparison of successions of virtual geomagnetic poles may reveal whether the geomagnetic field was predominantly dipolar (Creer 1970b; Creer & Ispir 1970) or non-dipolar (Cox 1969; Larson, Watson & Jennings 1971; and Hillhouse & Cox 1976) during a polarity transition. The polar paths defined by the virtual geomagnetic poles in different parts of the globe should be similar if the dipole field was predominant

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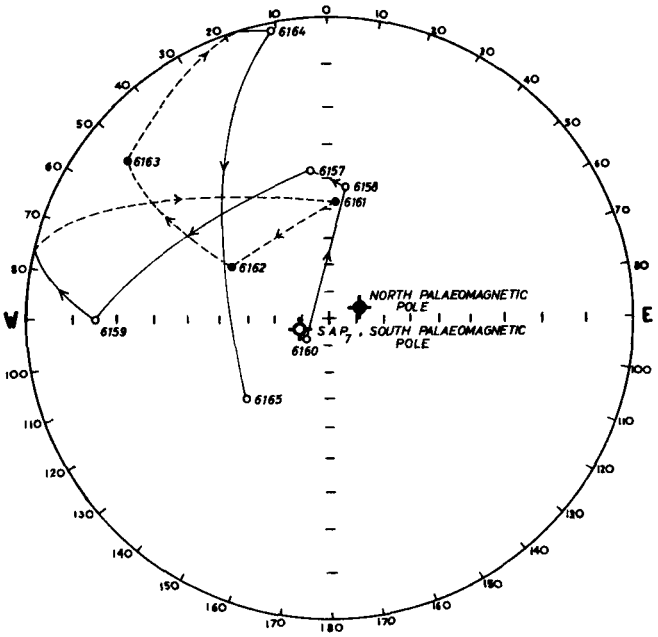


Figure 7. Virtual geomagnetic pole positions for the polarity transition recorded in the Middle Section of Paganzo Group exposures near Paganzo Village.

during the polarity transition; on the contrary the polar paths should be different if the non dipolar field was the predominant. The polarity transition recorded in the Middle Section of the Paganzo Group (Middle Permian, Quebrada del Pimiento Normal Magnetic Event) may offer an excellent opportunity to study the behaviour of the transitional geomagnetic field. Fig. 7 shows the succession of virtual geomagnetic poles for this polarity transition and we think it will be interesting to investigate other sequences of Middle Permian rocks from different parts of the globe.

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